FH Vorarlberg
University of Applied Sciences

Energy and Storage Systems - Trading

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Course Number 47083

Primary Energy Trading

- Limited availability and high demand
- Primary energy markets
  - Hard coal market
  - Natural gas market
  - Crude oil market
- No markets for uranium and lignite
  - Uranium: Strong political restrictions
  - Lignite: Not economical to transport (low energy / volume)
- Emission rights market
  - Consumer of fossil based primary energy resources
  - Deal with CO2 certificates

Markets for Electrical Power

- Electrical energy
- Reserve power (Dt. Regelenergie)
- Transmission line capacity (dt. Übertragungsnetzkapazitäten)

Trading of electrical energy

- Physical restrictions
  - transmission and distribution partly regulated

- Liberalized markets – Market participants
  - Production
  - Trading
  - Sales
  - Consumption

Different time requirements

- Good *electrical energy* can not be stored on larger scale
  - Variety of products

- Classified
  - Delivery period (dt. Lieferperiode)
  - Delivery profile (dt. Lieferprofil)
  - Trading period (dt. Handelszeitraum)
  - Settlement of transactions (dt. Erfüllung von Geschäften)

Product characteristic defines marketplace

- Product characteristics are defining the marketplace
- Base load profiles deliveries are not interrupted during delivery profile
- Peak load profiles are delivered during working days from 8.00 am – 8.00 pm

Marketplace	Delivery period/profile	Trading period
Intraday Trading	1h	1h to 1d prior delivery starts
Day Ahead Trading	1h	At 12.00 am the day prior delivery starts

Table 5: Product characteristics for spot trading (Intrady and Day-Ahead)

Product characteristic defines marketplace

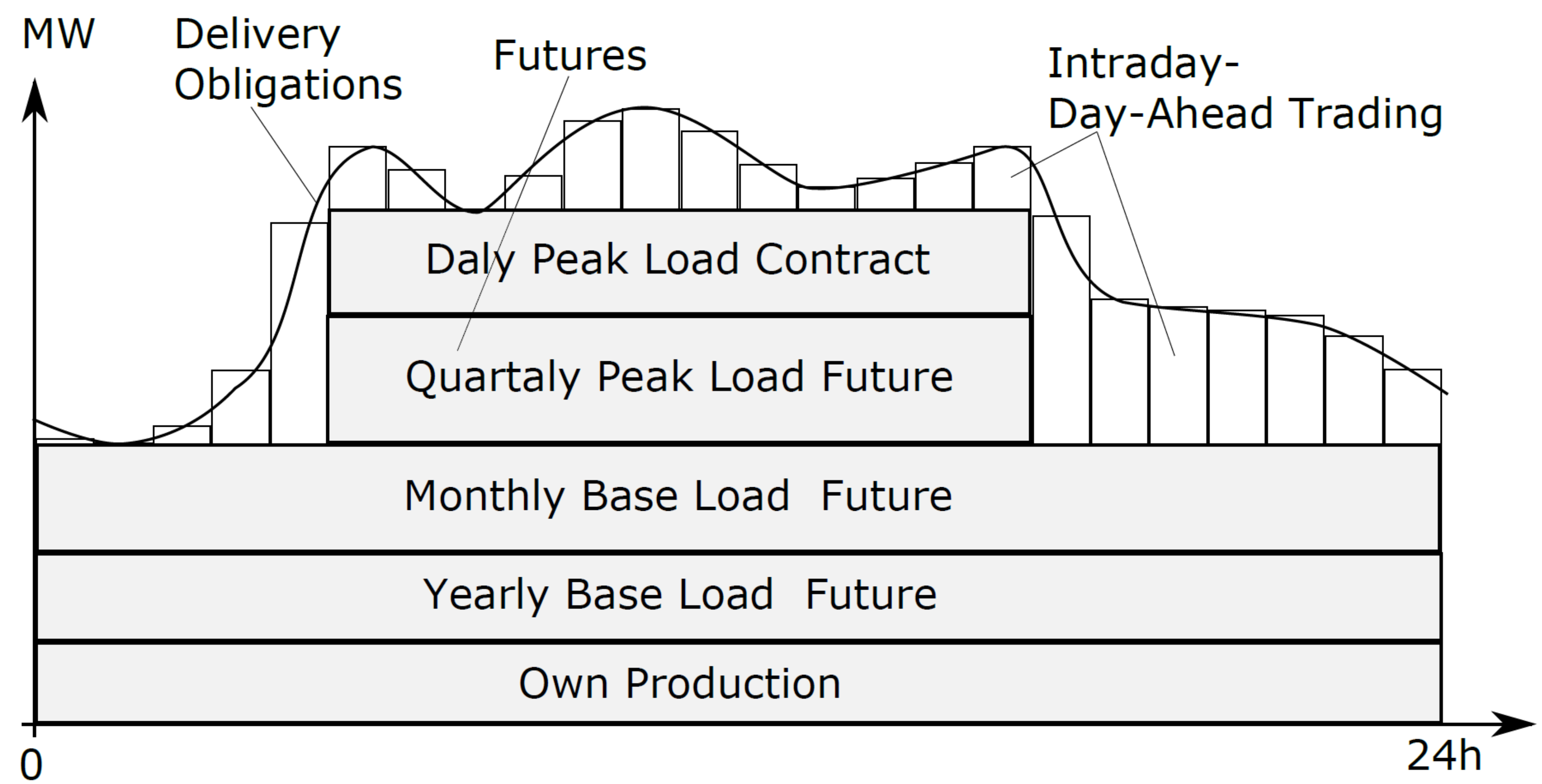
Delivery period/profile	Trading period
Weekly Base Load Weekly Peak Load	few days to four weeks prior delivery start
Monthly Base Load Monthly Peak Load	few days to six months prior delivery start
Quarterly Base Load Quarterly Peak Load	few days to seven quarters prior delivery start
Yearly Base Load Yearly Peak Load	few days to six years prior delivery start

Table 6: Product characteristics for futures trading (dt. Terminhandel)

Product characteristic defines marketplace

- Utilities are designing their optimal portfolios

- Portfolio – might be a mix of:
  - Own production
  - Contracts on spot markets
  - Contracts on future markets



Business Transactions – Continuous trading

- Two ways of doing business transactions
  - Continuous trading
  - Auctions
- Continuous trading
  - All buyer and seller orders are collected
  - Open ordering book – Everybody see non-executed orders
  - Order disappear (gets cleared) after business deal is finalized

Business Transaction – Continuous trading

- Order cleared if matching of
  - Quantity of electricity to a defined area
  - Price
- The price is never
  - higher than the purchase price fixed by the buyer
  - lower than the sales price offered by the seller
- Basic principle – Each order has a different price
- Markets using continuous trading
  - Intraday
  - Future markets

Business Action - Auctions

- All buying and selling orders are collected
- Closed ordering book
- Nobody can see available buy and sell options (Black box auction)
- As the auction starts
  - Fitting buying and selling orders are connected and executed
- Last executable deal defines the price for all deal done in this auction
  - Market Clearing Price, MCP (dt. Auktionspreis)
- Market using auctions
  - Day-Ahead trading

Business Action - Fulfillment

- Ways to fulfill business transactions
  - Physical fulfilling
  - Cash-Settlement
  - Cascading
  - Deliver of a future

Physical fulfilling

- Electrical energy is delivered physically
- To a defined destination
- Paid with money
- Usually done for
  - Intraday trading
  - Day-Ahead trading
  - Forwards and options in future trading

Cash-Settlement

- Usually used for
  - Monthly futures
- Business if fulfilled by money only
- Price is defined as the difference between
  - Price for the future itself
  - Average price of the underlying spot trading

Cascading

- Usually used for
  - quarterly and yearly futures
- Long term obligations are fulfilled by
  - several short term deliveries
- E.g. Quarterly base-load future is fulfilled by three monthly base-load futures

Delivery of a future

- This are options traded on stock markets
- They are fulfilled if defined future is delivered

Trading places

- Several market places evolved since liberalization
  - within an exchange (Dt. Börse)
  - outside an exchange

- Exchanges for electrical energy are highly regulated
- Participants and trading itself supervised by Government
  - Have to qualify to get access

- Exchange acts as central counterpart (dt. Zentraler Kontrahent)
  - Brings supply and demand orders together
  - Anonymous business transactions are possible

Trading outside exchange

- Outside an exchange
  - Called Over-The-Counter OTC trading
- OTC trading can be done via
  - Broker
  - Directly
- Brokers are similar to an exchange
  - Bring supply and demand orders together
  - Are service provider, they do not trade by themselves
  - Common to tell trading preferences
- Directly
  - OTC trading directly between buyer and seller

Power Exchange Trading Places

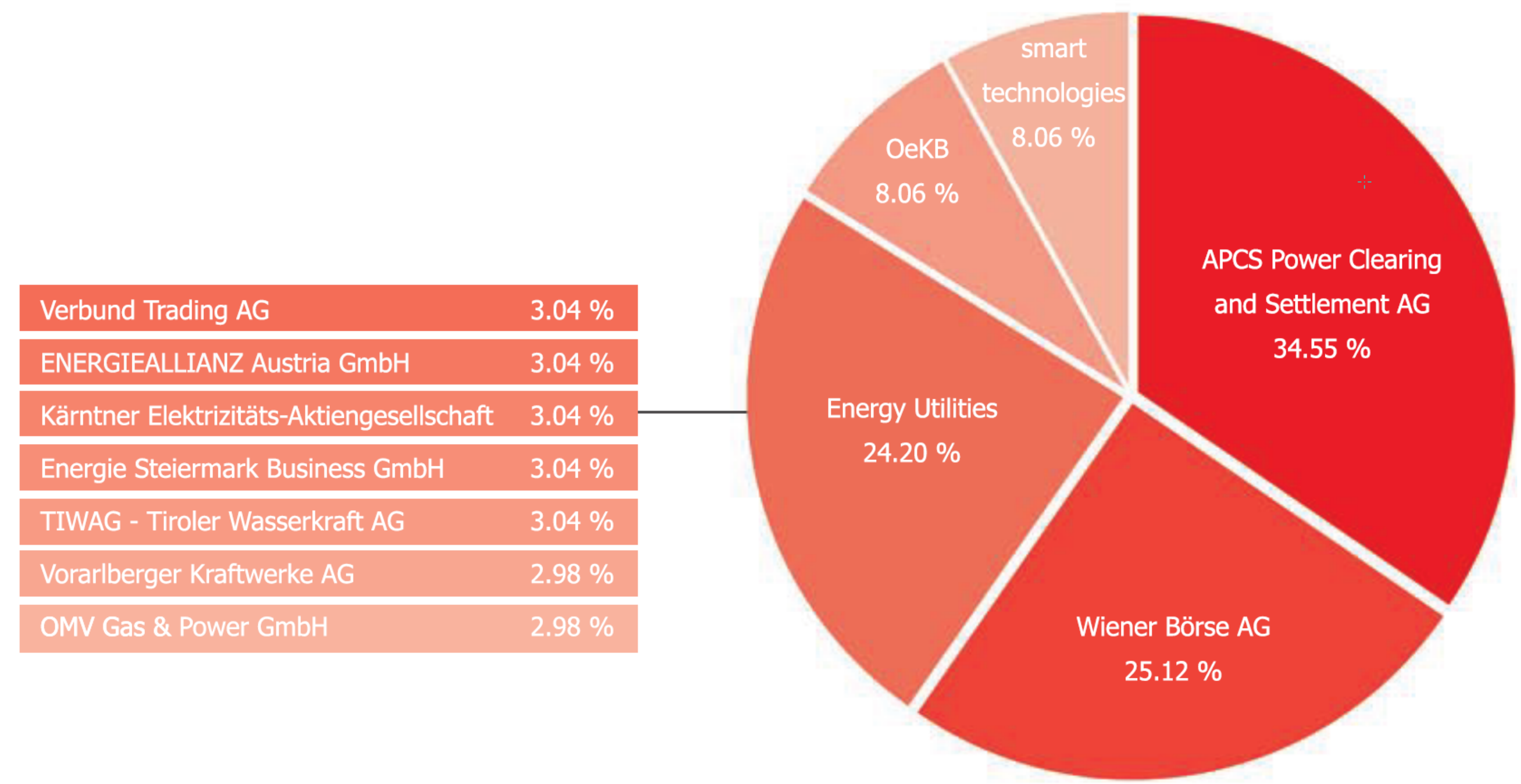
- Each trading place covers a certain area in Europe
- Where the business can physically be fulfilled

Name	Located	Product Type	Area
Energy Exchange Austria EXAA	Vienna	Spot Market	Austria, Germany
European Energy Exchange EEX	Leipzig	Futures Market	Germany, France
European Power Exchange EPEX Spot	Paris	Spot Market	Germany, France, Austria, Swiss
Nord Pool	Oslo	Spot Market	Norway, Sweden, Finland, Denmark, ...
NASDAQ OMX	Oslo	Futures Market	Norway, Sweden, Finland, Denmark, ...
Gestore Mercati Energetici GME	Rom	Futures Market Spot Market	Italy

Table 7: Important power exchange places in Europe

Most Important Trading Places

- Important Trading Places for Austrian's power industry
  - EXAA – Spot market
  - EEX – Future market
  - EPEX Spot – Spot market
- Energy Exchange Austria EXAA
  - Founded 2001 in Vienna
  - Spot market
  - For Germany and Austria
  - Trade volume 2015 8.22 TWh



Most Important Trading Places

- European Energy Exchange (EEX)
  - Founded 2002 in Leipzig, Germany
  - Futures trading (dt. Teminmarkt)

- European Power Exchange (EPEX Spot)
  - Founded 2008
  - Spot products
    - Day-Ahead
    - Intrady
  - Products can physically delivered to
    - Austrian, French, German or Swiss transmission system

Most Important Trading Places – EPEX Spot

- ... Continued
  - Trading volume 2014 382 TWh
  - Owned by EEX Group (51%) and HGRT (49%)
  - HGRT is composed of the involved TSO's
    - Amprion
    - APG
    - Elia
    - RTE
    - Swissgrid
    - TenneT

Most Important Trading Places – EPEX Spot

- Counterpart at EPEX
  - European Commodity Clearing ECC
- ECC gets the transferred trades
  - which nominates exchange of energy to the TSO of the delivery area
  - makes sure the money gets transferred to the seller

OTC Trading

- Most of the business (70% in Germany)
- Important brokers for Europe
  - Spectron Group Limited
  - Tradition Financial Servies
  - Tullett Prebon Group Limited
  - ICAP plc
  - GFI Group Inc.

Day Ahead Trading

- Used to optimize the electrical energy portfolio on a daily base
- Done for each quarterly hour on the upcoming day

- Participant calculates the needed energy to fulfill obligations
  - Missing energy – buy order
  - Surplus energy – sell order

- Put buy/sell orders into the Day Ahead Trading system of the EPEX

Day Ahead Trading

- Put orders into the Day Ahead Trading system of the EPEX

- Forecast shows more power required
  - Increase own power production
  - Buy missing amount

- Prefer to buy if the generation costs epsilon are higher than the price on the market

Example

- Obligations of 30 MW
- Own production costs
  - Block 1 – 20 €/MWh
  - Block 2 – 40 €/MWh
  - Block 3 – 60 €/MWh
  - Block 4 – 80 €/MWh

- Scenario 1 – Block 1-3 are producing and Block 4 is turned off
- Scenario 2 – Block 2-3 are producing. Block 3-4 are turned off. Missing Energy is bought via Day Ahead order for 48 €/MW

- Average production cost for Scenario 1 and Scenario 2?

Example

- Average production cost for Scenario 1 – 40 €/MWh
- Average production cost for Scenario 2 – 36 €/MWh

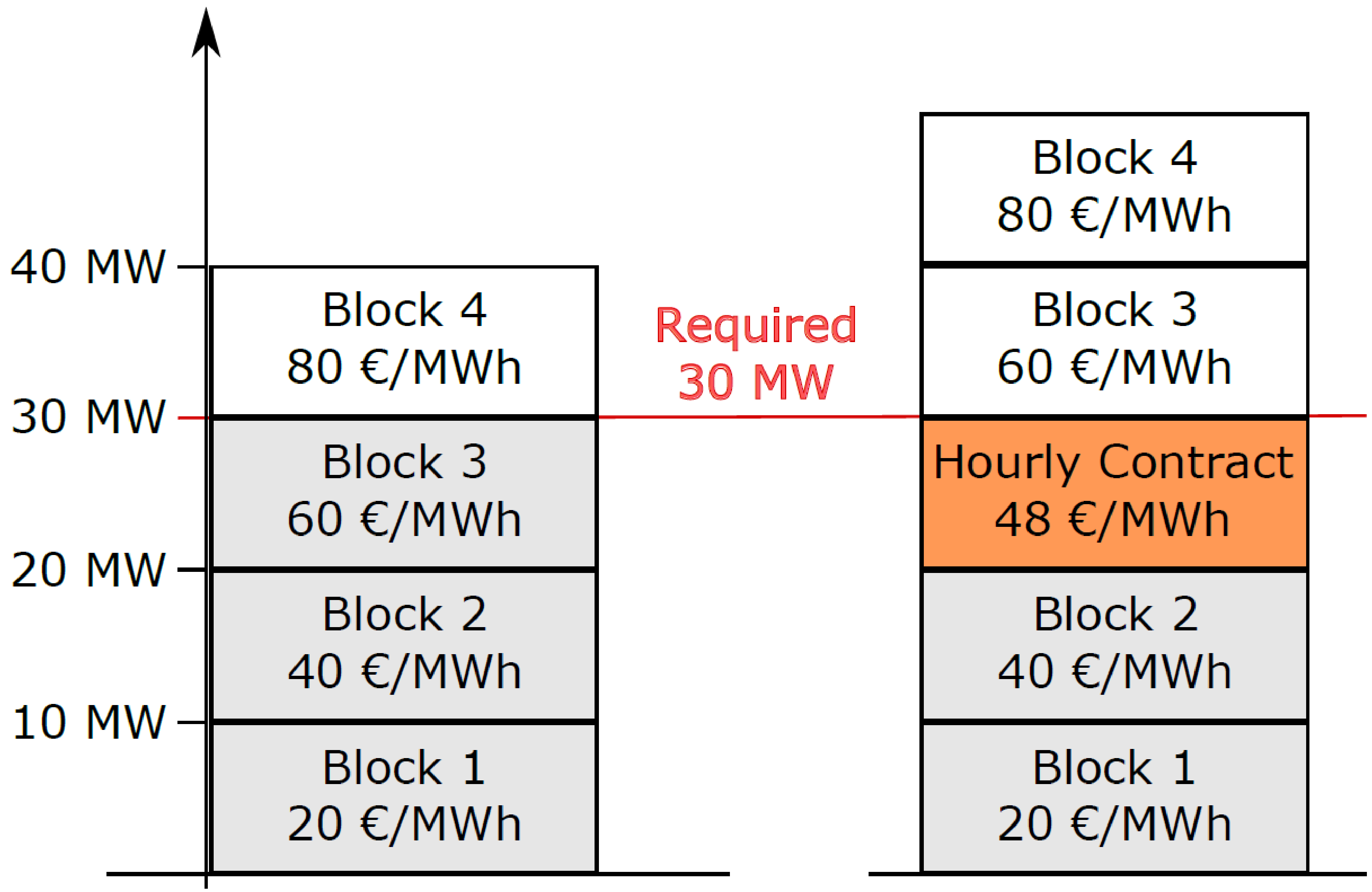


Figure 25: Portfolio optimization example.

Day Ahead Trading

- Orders are submitted
  - limit for the purchase price or
  - limit for the sales price

- Each order is a combination
  - price €/MWh
  - quantity MWh for a certain time on the following day

Day Ahead Trading

- Orders are submitted via web frontend

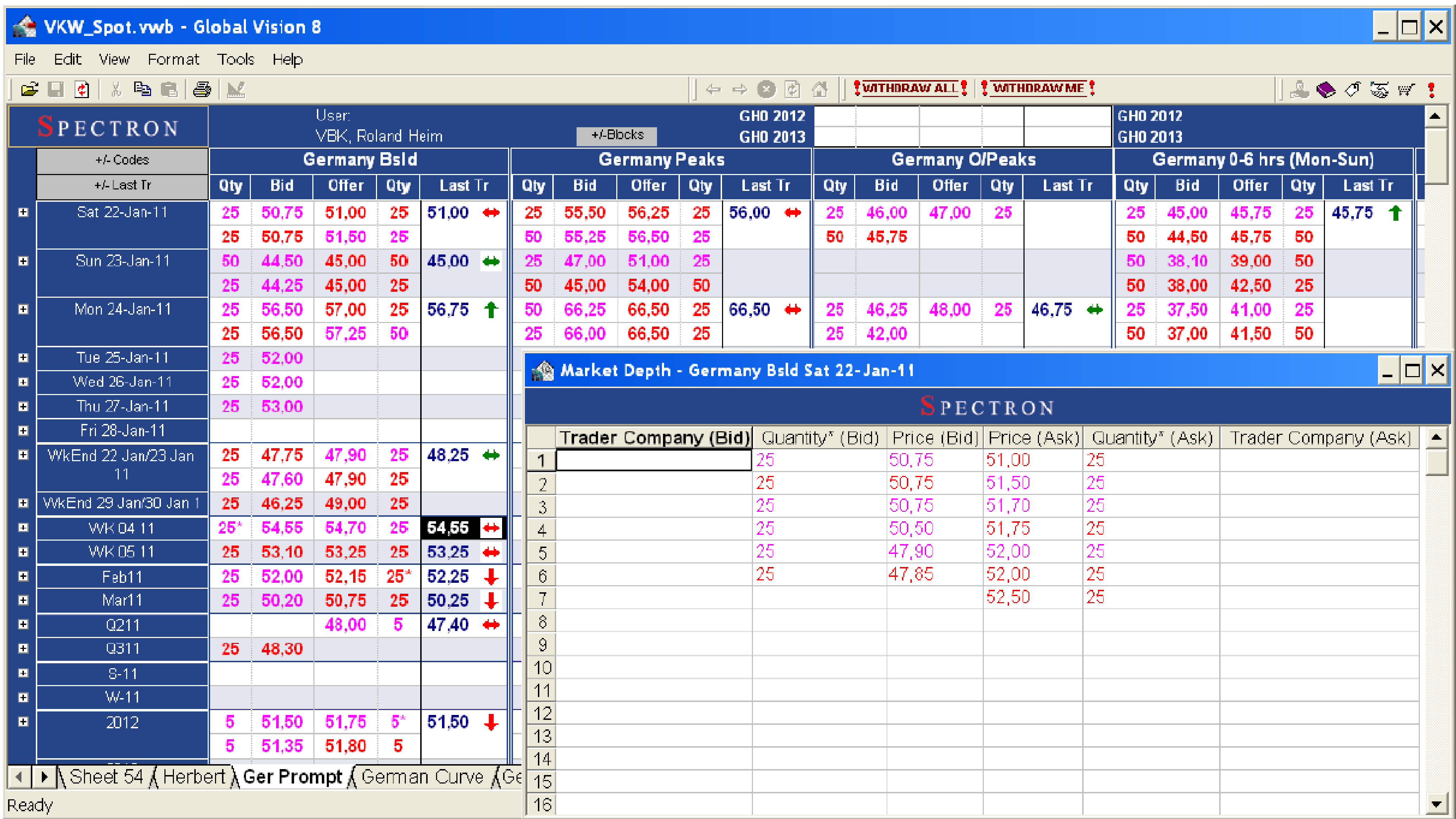
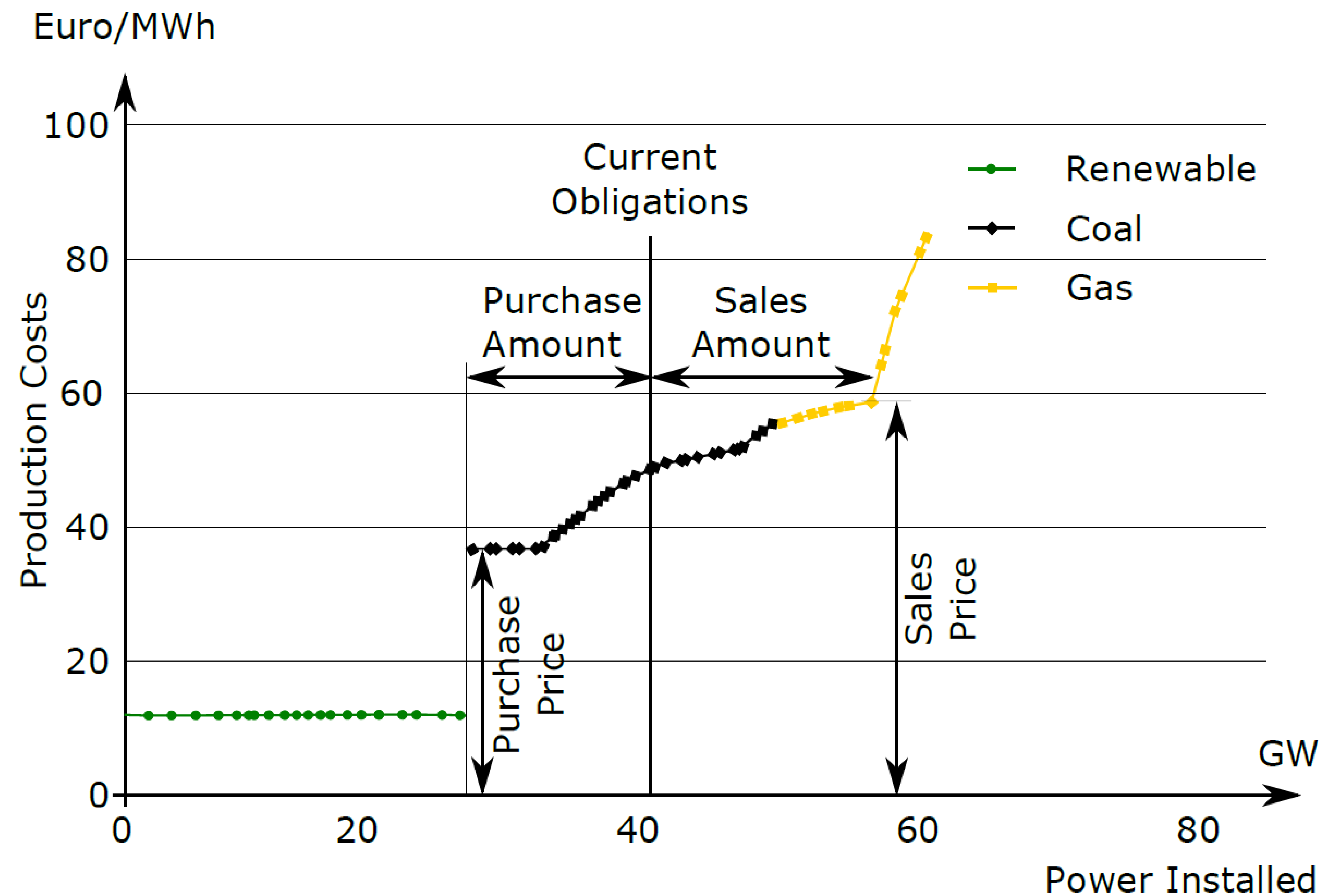


Figure 26: Energy trading frontend
Source: Schmöller and Preuschoff (2016)

Merit Order Curve

- Power producer gets its cost from
  - Merit Order Curve
- Merit Order Curve
  - Ranking all power sources
  - Ascending order of price / costs of production together with produced amount of energy
- Left side
  - Usually renewables with very low production costs
  - Nuclear power plants
  - Lignite power plants



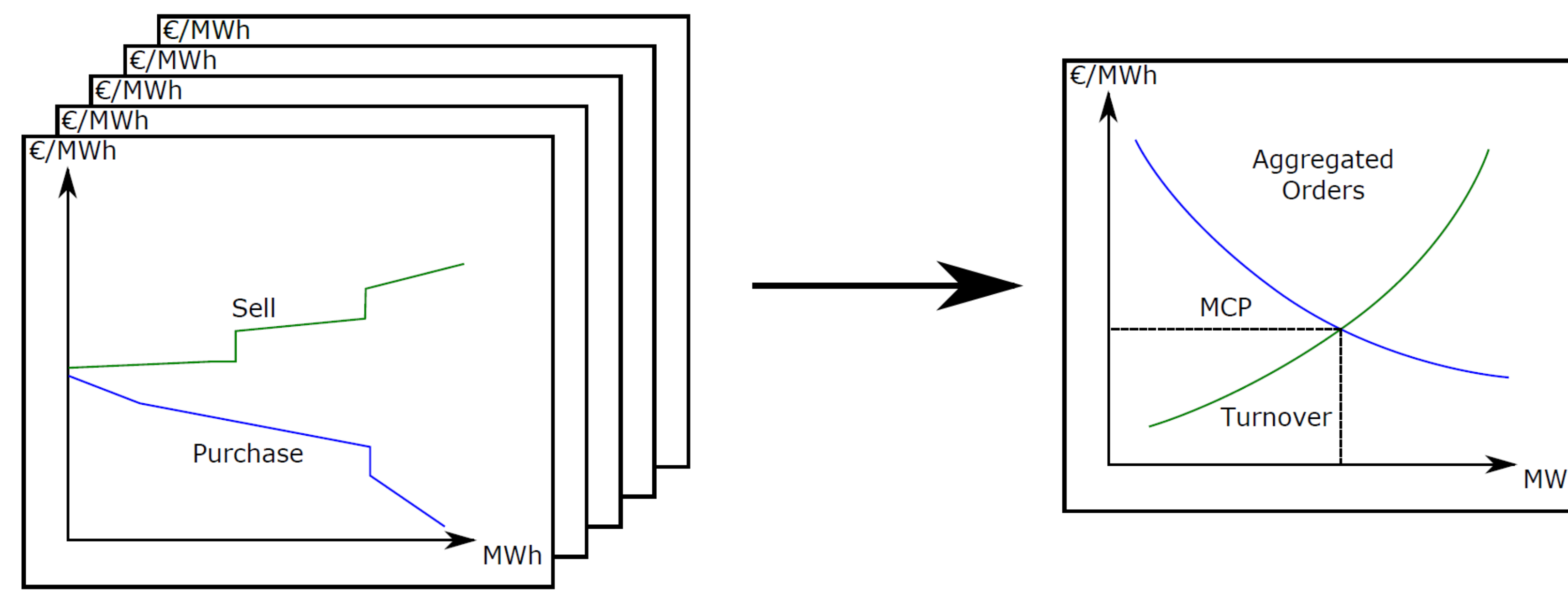
Merit Order Curve

... continued

- Right side
  - Expensive peak load power plants
  - Pump storage
  - Gas turbines

Day Ahead Trading

- Price determined
  - after all orders are submitted (usually 12.00 am)
  - price for each time unit of the next day is determined
  - Therefore all orders are aggregated
  - Result: two curves
  - Intersection defines the price
  - Intersection point: Market clearing price MCP for a particular time unit of the following day
- Common price for all Day Ahead trading participants



Day Ahead Trading

... continued

- Only selling orders within the defined limits are executed

- Example
  - My purchase limit 25 €/MWh
  - MCP 40 €/MWh
  - -> I will not purchase because it is cheaper to produce by myself (25 €/MWh)

  - My selling limit is 60 €/MWh
  - MCP 40 €/MWh
  - -> I will not sell because my production cost is higher than what I can earn

Group Orders

- It is common practice to group orders to a block
- Group is only executed
  - when all individual's are executed
- Group is not executed
  - if one of the group is not executed

- Bloc orders are used to ensure that a power plant is not started up to fulfill an order for only a short period of time. E.g. one hour

Phelix-Index

- Physical Electricity Index (Phelix)
- Average value of the MCP's for the following day
- Phelix Base
  - Average value for the entire day
- Phelix Peak
  - Average value for delivery hour 9-20, peak load

Intraday Trading

- Day-Ahead trading is based on production and consumption forecast
- Always deviation between them
  - Real consumption different
  - Fluctuating primary energies (wind and sun)
  - Unexpected break down of a generation unit

- Therefore – Intraday Trading
  - Trading within a day
  - Continuous trading with an open ordering book

Futures Trading (Dt. Terminhandel)

- Used to compensate for the volatile prices at the spot markets
- Future
  - Commitment to trade
    - certain amount of energy
    - clear defined price
    - at a point in the future
- Time frames
  - weeks, months, quarters or years
- Help to create long term plans for the bulk of available energy
- Business if fulfilled via cash settlements

Trading of balancing power

- Control area manager (in Austria APG) is responsible to keep power system balanced
  - power system voltage
  - power system frequency

- To do that they need reserve power
- Reserve power market provides this energy

- High technical requirements
- Participants need to qualify first

Trading of transmission line capacity

- Transmission lines are needed to fulfill trading contracts
- Transmission system operator TSO is in charge
- Does auctions where the capacities can be purchased

Balancing groups

- Balancing groups are needed so consumers, producers, wholesaler can conclude deals with each other
- Each grid user (consumer or producer) concludes a contract with an grid operator and another with the desired supplier or trader
- Whoever takes electricity
  - off the grid
  - feed in
  - trades

  must be a member of a balance group

Balancing groups

- Central clearing and settlement agent CSA
  - Dt. Bilanzgruppenkoordinator
  - In Austria: Austrian Power Clearing & Settlement (APCS)
- CSA gets the schedules from the balance group and the metering data from the consumer and supplier
- Balance responsible parties BRP
  - Dt. Bilanzgruppenverantwortlicher

  are registered at the APCS

Balance Group System in Austria

- Examples for balance responsible parties registered at APCS
  - Vorarlberger Kraftwerke AG
  - Vattenfall Energy Trading
  - TIWAG – Tiroler Wasserkraft
  - RWE Supply & Trading
  - ÖBB Infrastruktur
  - Next Krafwerke AG
  - OeMAG, Öko-balance group representative

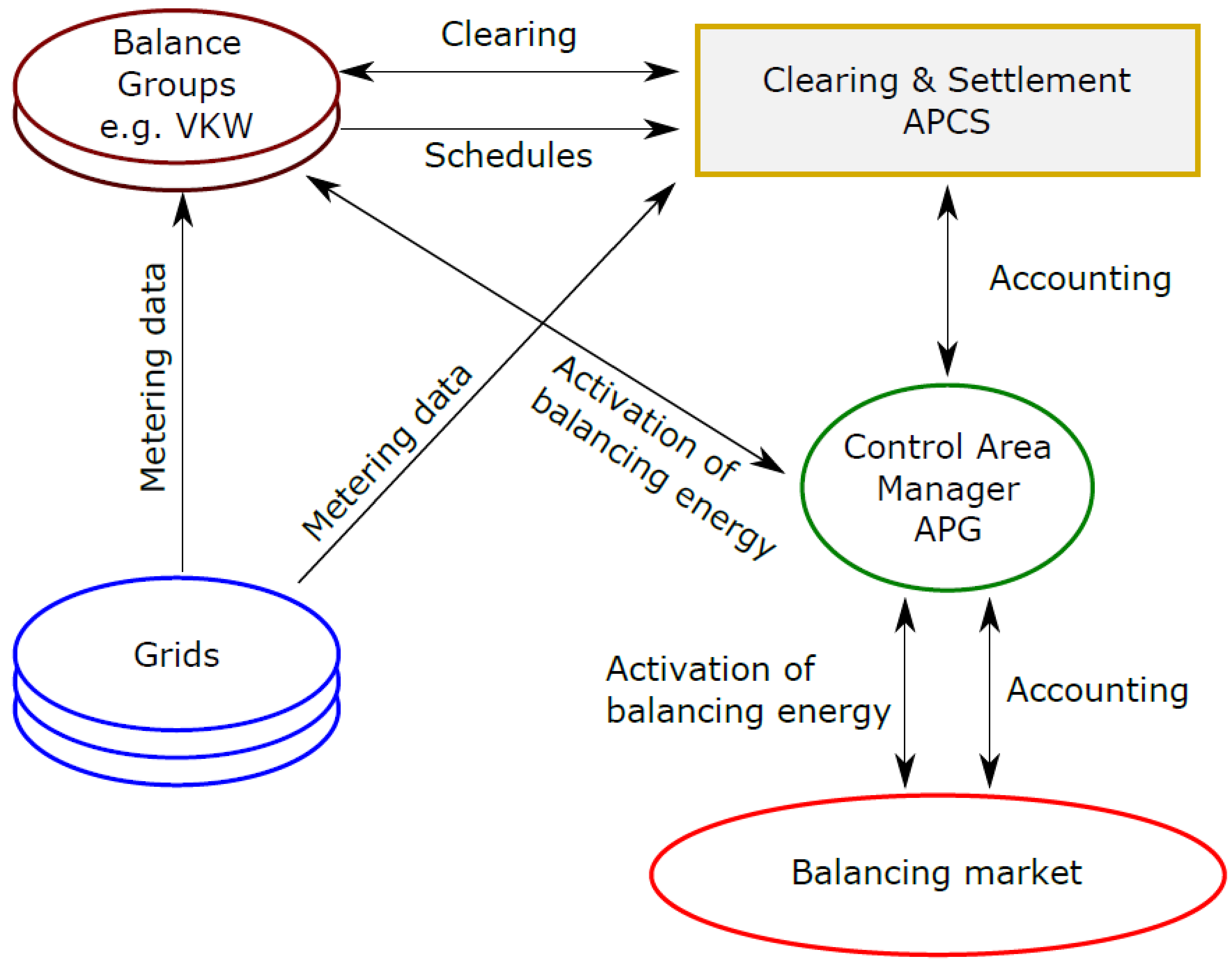


Figure 29: Balance Group System in Austria

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